

Insert uppercase

SQL

```
INSERT INTO student(id, name, city)
VALUES (1, UPPER('tarun'), UPPER('indore'));
```

Insert lowercase

SQL

```
INSERT INTO student(id, name, city)
VALUES (2, LOWER('RAHUL'), LOWER('BHOPAL'));
```

1. Numeric Data Types

Data Type	Size	Range / Use
TINYINT	1 byte	-128 to 127
SMALLINT	2 bytes	-32,768 to 32,767
MEDIUMINT	3 bytes	-8M to 8M
INT / INTEGER	4 bytes	-2B to 2B
BIGINT	8 bytes	बहुत बड़े integers
FLOAT	4 bytes	Approximate decimal
DOUBLE	8 bytes	More precision
DECIMAL(M,D)	Variable	Exact decimal values



2. String Data Types

Data Type	Size	Use
CHAR(n)	Fixed	Fixed-length text
VARCHAR(n)	Variable	Variable-length text
TINYTEXT	Up to 255 bytes	Short text
TEXT	Up to 65,535 bytes (≈64 KB)	Paragraphs
MEDIUMTEXT	Up to 16 MB	Large text
LONGTEXT	Up to 4 GB	Very large text

3. Binary Data Types

Data Type	Use
<code>BINARY(n)</code>	Fixed binary data
<code>VARBINARY(n)</code>	Variable binary data
<code>TINYBLOB</code>	Up to 255 bytes
<code>BLOB</code>	Up to 64 KB
<code>MEDIUMBLOB</code>	Up to 16 MB
<code>LOBLOB</code>	Up to 4 GB

BLOB का उपयोग images, PDFs, audio जैसी binary files के लिए किया जाता है।

4. Date & Time Data Types

Data Type	Size	Example
<code>DATE</code>	3 bytes	<code>2026-07-13</code>
<code>TIME</code>	3 bytes	<code>14:30:45</code>
<code>YEAR</code>	1 byte	<code>2026</code>
<code>DATETIME</code>	5 bytes	<code>2026-07-13 14:30:45</code>
<code>TIMESTAMP</code>	4 bytes	Auto timestamp

7. ENUM Data Type

केवल predefined values store करता है।

SQL

```
gender ENUM('Male', 'Female', 'Other')
```

8. SET Data Type

एक से अधिक predefined values store कर सकता है।

SQL

```
skills SET('C++', 'Java', 'Python', 'SQL')
```

9. Spatial Data Types (GIS)

Data Type	Use
POINT	Single location
LINestring	Road/Path
POLYGON	Area
GEOMETRY	Generic geometry

1-byte character set (Interview के लिए आसान table)

Data Type	Maximum Characters	Maximum Bytes 
CHAR(255)	255	255 bytes
VARCHAR	लगभग 65,535 characters*	लगभग 65,535 bytes*
TINYTEXT	255	255 bytes
TEXT	65,535	65,535 bytes ($\approx$ 64 KB)
MEDIUMTEXT	16,777,215	16,777,215 bytes ($\approx$ 16 MB)
LONGTEXT	4,294,967,295	4,294,967,295 bytes ($\approx$ 4 GB)

Data Type	Max Size
TINYTEXT	255 B
TEXT	64 KB
MEDIUMTEXT	16 MB
LONGTEXT	4 GB

```

CREATE TABLE student (
    id INT PRIMARY KEY,

    phone VARCHAR(10),
    CHECK (phone REGEXP '^[6-9][0-9]{9}$'),

    aadhaar CHAR(12),
    CHECK (aadhaar REGEXP '^[0-9]{12}$'),

    dob VARCHAR(10),
    CHECK (dob REGEXP '^(0[1-9]|12)[0-9]|3[01])/(0[1-9]|1[0-2])/[0-9]{4}$'),

    password VARCHAR(16),
    CHECK (
        CHAR_LENGTH(password) = 16
        AND password REGEXP '[A-Z]'
        AND password REGEXP '[a-z]'
        AND password REGEXP '[0-9]'
        AND password REGEXP '[@#]$'
    )
);

```



```

CHECK (
    CHAR_LENGTH(password)=16
    AND password REGEXP '[A-Z]'
    AND password REGEXP '[a-z]'
    AND password REGEXP '[0-9]'
    AND password REGEXP '[@#]$'
)

```

Sql_insert_part_1